

AMMONIA (ANHYDROUS)**ICSC: 0414 (October 2013)**

R717

Refrigerant gas 717

CAS #: 7664-41-7**UN #: 1005****EC Number: 231-635-3**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Flammable. Cylinder may explode in heat of fire. Gas/air mixtures are explosive.	NO open flames, NO sparks and NO smoking. Closed system, ventilation, explosion-proof electrical equipment and lighting.	In case of fire in the surroundings, use appropriate extinguishing media. In case of fire: keep cylinder cool by spraying with water.

AVOID ALL CONTACT! IN ALL CASES CONSULT A DOCTOR!

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Burning sensation. Cough. Laboured breathing. Shortness of breath. Sore throat.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Half-upright position. Administration of oxygen may be needed. Refer immediately for medical attention.
Skin	Redness. Pain. Blisters. Skin burns. ON CONTACT WITH LIQUID: FROSTBITE.	Cold-insulating gloves. Protective clothing.	Rinse skin with plenty of water or shower for at least 15 minutes. ON FROSTBITE: rinse with plenty of water, do NOT remove clothes. Refer immediately for medical attention.
Eyes	Redness. Pain. Severe burns. ON CONTACT WITH LIQUID: FROSTBITE.	Wear face shield or eye protection in combination with breathing protection.	Rinse with plenty of water for several minutes (remove contact lenses if easily possible). Refer immediately for medical attention.
Ingestion			

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Evacuate danger area! Consult an expert! Personal protection: gas-tight chemical protection suit including self-contained breathing apparatus. Ventilation. Shut off cylinder if possible. Isolate the area until the gas has dispersed. Remove gas with fine water spray. NEVER direct water jet on liquid.	According to UN GHS Criteria  DANGER Flammable gas Contains gas under pressure; may explode if heated Toxic if inhaled Causes severe skin burns and eye damage Very toxic to aquatic life Transportation UN Classification UN Hazard Class: 2.3; UN Subsidiary Risks: 8
STORAGE	
Fireproof. Separated from oxidants, acids and halogens. Cool. Keep in a well-ventilated room.	
PACKAGING	



Prepared by an international group of experts on behalf of ILO and WHO, with the financial assistance of the European Commission.
© ILO and WHO 2021



AMMONIA (ANHYDROUS)**ICSC: 0414****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

COLOURLESS GAS OR COMPRESSED LIQUEFIED GAS WITH PUNGENT ODOUR.

Physical dangers

The gas is lighter than air.

Chemical dangers

Mixtures with mercury, silver and gold oxides are shock-sensitive. The substance is a strong base. It reacts violently with acid and is corrosive. Reacts violently with strong oxidants, halogens and many other substances. Attacks copper, aluminium, zinc and their alloys. Dissolves in water evolving heat. The substance reacts with most organic and inorganic compounds, causing fire and explosion hazard.

Formula: NH_3

Molecular mass: 17.0

Boiling point: -33°C

Melting point: -78°C

Relative density (water = 1): 0.7 (-33°C)

Solubility in water, g/100ml at 20°C : 54

Vapour pressure, kPa at 26°C : 1013

Relative vapour density (air = 1): 0.60

Auto-ignition temperature: 630°C

Explosive limits, vol% in air: 15-33.6

EXPOSURE & HEALTH EFFECTS**Routes of exposure**

The substance can be absorbed into the body by inhalation.

Effects of short-term exposure

Rapid evaporation of the liquid may cause frostbite. The substance is corrosive to the eyes, skin and respiratory tract. Exposure could cause asphyxiation due to swelling in the throat. Inhalation may cause lung oedema, but only after initial corrosive effects on eyes and/or airways have become manifest.

Inhalation risk

A harmful concentration of this gas in the air will be reached very quickly on loss of containment.

Effects of long-term or repeated exposure

Repeated or chronic inhalation of the vapour may cause chronic inflammation of the upper respiratory tract. Lungs may be affected by repeated or prolonged exposure. This may result in chronic obstructive pulmonary disorders (COPD).

OCCUPATIONAL EXPOSURE LIMITS

TLV: 25 ppm as TWA; 35 ppm as STEL.

EU-OEL: 14 mg/m^3 , 20 ppm as TWA; 36 mg/m^3 , 50 ppm as STEL.

MAK: 14 mg/m^3 , 20 ppm; peak limitation category: I(2); pregnancy risk group: C

ENVIRONMENT

The substance is very toxic to aquatic organisms. It is strongly advised not to let the chemical enter into the environment.

NOTES

Ammonia is normally supplied in compressed liquified form in cylinders.

See ICSC 0215.

Turn leaking cylinder with the leak up to prevent escape of gas in liquid state.

ADDITIONAL INFORMATION**EC Classification**

Symbol: T, N; R: 10-23-34-50; S: (1/2)-9-16-26-36/37/39-45-61; Note: U

All rights reserved. The published material is being distributed without warranty of any kind, either expressed or implied. Neither ILO nor WHO nor the European Commission shall be responsible for the interpretation and use of the information contained in this material.